

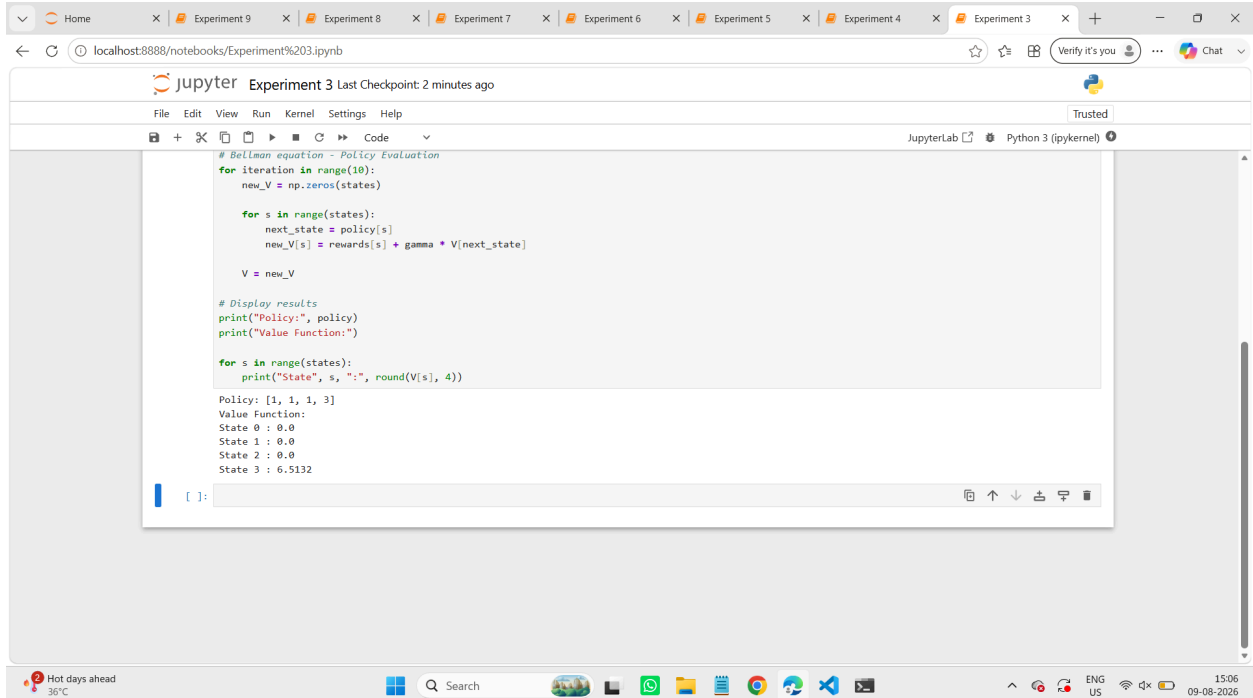
# Experiment 3 RL

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Course Code: MLA0305

Slot: B



```
# Bellman equation - Policy Evaluation
for iteration in range(10):
    new_V = np.zeros(states)

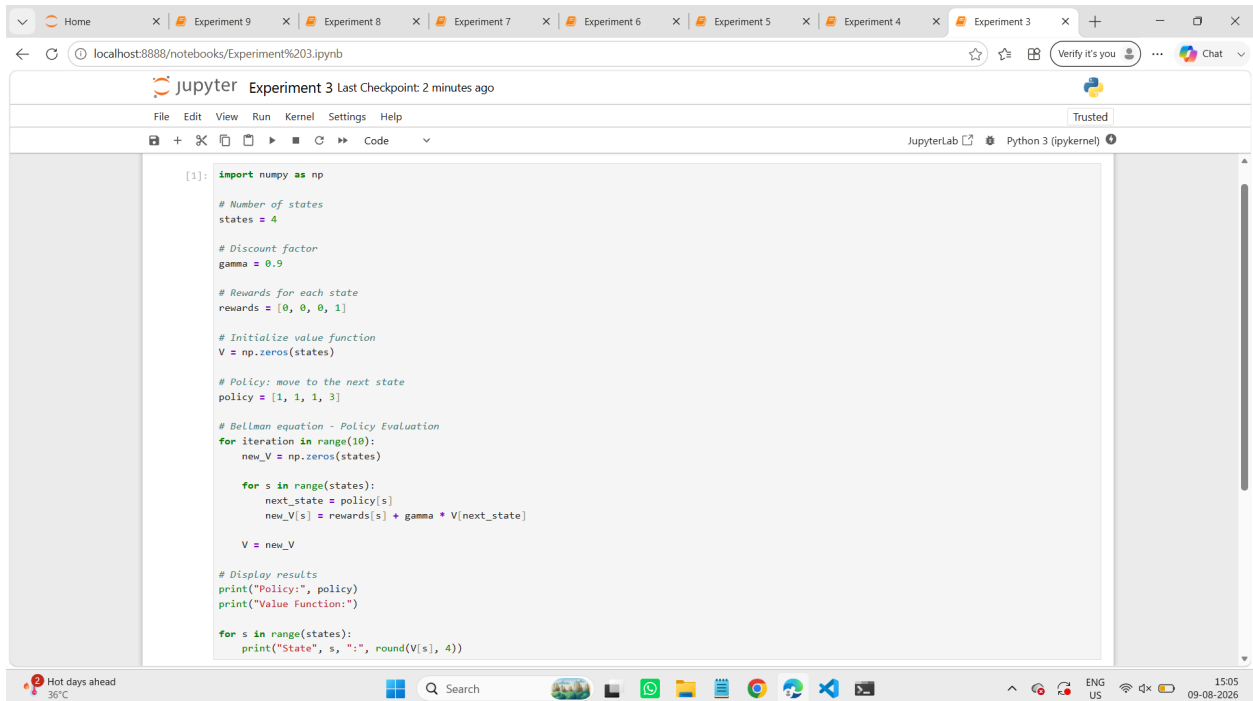
    for s in range(states):
        next_state = policy[s]
        new_V[s] = rewards[s] + gamma * V[next_state]

    V = new_V

# Display results
print("Policy:", policy)
print("Value Function:")

for s in range(states):
    print("State", s, ":", round(V[s], 4))

Policy: [1, 1, 1, 3]
Value Function:
State 0 : 0.0
State 1 : 0.0
State 2 : 0.0
State 3 : 6.5132
```



```
[1]: import numpy as np

# Number of states
states = 4

# Discount factor
gamma = 0.9

# Rewards for each state
rewards = [0, 0, 0, 1]

# Initialize value function
V = np.zeros(states)

# Policy: move to the next state
policy = [1, 1, 1, 3]

# Bellman equation - Policy Evaluation
for iteration in range(10):
    new_V = np.zeros(states)

    for s in range(states):
        next_state = policy[s]
        new_V[s] = rewards[s] + gamma * V[next_state]

    V = new_V

# Display results
print("Policy:", policy)
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for s in range(states):
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